B9 Energy, Springs and Materials

Give numeric answers in SI form, without prefixes. Assume extension is proportional to tension.

B9.1 A spring of natural length 12.2 cm is stretched to 14.7 cm when a force of 8.0 N is applied.

a) How much work is done on the spring to stretch it?

b) How much elastic potential energy (EPE) is stored in the spring?

B9.2 Calculate how much extra work must be done in order to stretch a spring from 12.2 cm to 13.2 cm, if its spring constant is 15 N cm^{-1} and natural length 10.2 cm.

B9.3 A wire of natural length 2.508 m, cross sectional area 1.0 mm^2 and Young's modulus 80 GPa is stretched to a new length of 2.512 m, which is below the limit of proportionality. How much work was done in order for this to happen?

Answers

B9.1

a) $0.5 \times 8 \times 0.025$ =0.100 J

b) Equal to last part

B9.2 $0.5 \times 1500 \times \left(0.03^2 - 0.02^2\right)$ =0.375 J

B9.3 $F = \sigma A = E\epsilon A = 80 \times 10^9 \times \dfrac{0.004}{2.508} \times 1 \times 10^{-6}$ =127.59 N

Work = $0.5 \times 127.59 \times 0.004$ =0.25518 J